MAT 3103: Computational Statistics and Probability **Semester: Fall 2019-20**

Chapter 8: Test of Hypothesis

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Parameter: It is the unknown constant characteristic of the population observations.

Example: If $X \sim N(\mu, \sigma^2)$, then μ and σ^2 are parameters.

Statistic: It is the function of sample observations.

Example: If $X \sim N(\mu, \sigma^2)$, then $\bar{x} = \frac{\sum x_i}{n}$, $s^2 = \frac{1}{n-1}[\sum x_i^2 - \frac{(\sum x_i)^2}{n}]$ are statistics.

Hypothesis: Any statement regarding one or more of parameters of a population or about population is called hypothesis.

Null hypothesis (H_0): Any statement regarding parameter of a population or population which is used for possible rejection by test is called null hypothesis.

Alternative hypothesis (H_1): Any statement against null hypothesis is alternative hypothesis.

Example: Given, test scores of 2 random samples, does one group differ from other?

A possible null hypothesis, $H_0 : \mu_1 = \mu_2$, alternative hypothesis, $H_1 : \mu_1 \neq \mu_2$. [μ_1 = mean of population 1 and μ_2 = mean of population 2]

Acceptance region: If the value of the test statistic falls into the probability space of the distribution of the test statistic and lead us to accept the null hypothesis then the probability space is called the acceptance region.

Critical Region: The probability space in which the test statistic falls and leads us to reject the null hypothesis is called critical region or rejection region. In the given figure below the critical and acceptance region are shown.



If $Z_{cal} > Z_{\alpha/2}$, then H_0 is rejected [cal=calculated value & $Z_{\alpha/2} = Z_{tab}$=Tabulated value]

If $Z_{cal} < Z_{\alpha/2}$, then H_0 cannot be rejected.

Level of significance: It is the probability of rejecting the null hypothesis when the null hypothesis is true. We denote it by α, usually $\alpha = 0.05$.

Degrees of freedom (df): The number of independent values which are used in deriving a test statistic is known as degrees of freedom (df).

Test of hypothesis: The statistical procedure to verify the null hypothesis using any test statistic is known as test of hypothesis.

Test statistic: The function of sample observations which is used to verify the null hypothesis is called test statistic.

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}},$$

is a test statistic used for testing mean.

Types of tests:

- Parametric test
- Non-parametric test

Parametric test: Conventional statistical procedures are also called parametric tests. In a parametric test a sample statistic is obtained to estimate the population parameter. Because this estimation process involves a sample, a sampling distribution, and a population, certain parametric assumptions are required to ensure all components are compatible with each other. For example, in Analysis of Variance (ANOVA) there are three assumptions:

- Observations are independent.
- The sample data have a normal distribution.
- Scores in different groups have homogeneous variances.

Example: t-test, z-test, F-test etc.

Non-parametric test: Non-parametric tests do not make parametric assumptions. It focus on order or ranking.

Example: Chi-square (χ^2) test (for categorical fields), Kolmogorov-Smirnov test (for continuous fields)

Statistical reference is concerned two types of problem:

- Estimation of population parameters
- Tests of hypothesis

When the data under analysis are met those assumptions for parametric tests, we should choose parametric tests because they are more powerful than non-parametric tests.

8.1 Different examples of Parametric & Non-parametric tests

8.1.1 Test regarding mean:

Let $x_1, x_2, \ldots, x_n$ be n sample observations drawn independently from a population with mean μ and variance σ^2. Let us assume that the sample observations follow normal distribution, i.e. $x \sim N(\mu, \sigma^2)$. The problem is to test the null hypothesis $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.

Assumptions:

1. σ^2 is unknown and n is small ($n < 30$)
2. σ^2 is unknown and n is large ($n \geq 30$)
3. σ^2 is known (n is small or large) and its value is, say σ_0^2

Under assumption 1, the test statistic to test the significance of H_0 is,

$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}} \sim t_{n-1}$ [n=Sample size]

Under assumption 2, the test statistic to test the significance of H_0 is,

$Z = \frac{\bar{x}-\mu_0}{s/\sqrt{n}} \sim N(0,1) = Z_{\alpha/2} = Z_{0.05/2} = Z_{0.025} = 1.96$

Under assumption 3, the test statistic to test the significance of H_0 is,

$Z = \frac{\bar{x}-\mu_0}{\sigma_0/\sqrt{n}} \sim N(0,1) = Z_{\alpha/2} = Z_{0.05/2} = Z_{0.025} = 1.96$

8.1.1.1 Test regarding one sample mean

Example 1: A random sample of 10 persons is selected and their level of education (in completed years of schooling) is recorded. The sample observations are, x: 5, 2, 0, 4, 16, 14, 10, 10, 6, 5. Do you think that the average schooling years of the persons in population is 5?

Solution:

Let assume that $x \sim N(\mu, \sigma^2)$, where σ^2 is unknown and $n = 10$ [$n < 30$].

Hypothesis:

Null hypothesis, $H_0 : \mu = \mu_0 = 5$

Alternative hypothesis, $H_1 : \mu \neq \mu_0$

Test statistics, $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$.....(1)

Now, $\bar{x} = \frac{\sum x_i}{n} = 72/10 = 7.2$

$s^2 = \frac{1}{n-1}[\sum x_i^2 - \frac{(\sum x_i)^2}{n}] = 1/9(758 - 72^2/10) = 23.96$, $s = \sqrt{23.96} = 4.89$

From equation (1), $t = \frac{7.2-5}{4.89/\sqrt{10}} = 1.42$

$t_{cal} = |t| = 1.42$ and $t_{tab} = t_{n-1} = t_9 = 2.262$ [from t-table with 5% level of significance]

$t_{cal} < t_{tab}$, so we cannot reject H_0 and the average schooling years of the persons in population is 5.

8.1.1.2 Test of equality of two means

Example 2:

University	Number of graduates employed in computer related job
A	18, 16, 15, 20, 18, 15, 12
B	20, 14, 12, 22, 16, 14, 15, 10, 12, 18, 10

Do you think that the employment facility for both the universities is similar?

Solution:

Let x_1 be the number of graduates of university A and x_2 be the number of graduates of university B. Assume that $x_1 \sim N(\mu_1, \sigma_1^2)$, $x_2 \sim N(\mu_2, \sigma_2^2)$.

Also assume that $\sigma_1^2 = \sigma_2^2 = \sigma^2$ and it's unknown. Sample size, $n_1 = 7$, $n_2 = 11$ (< 30).

Hypothesis:

Null hypothesis, $H_0 : \mu_1 = \mu_2$

Alternative hypothesis, $H_1 : \mu_1 \neq \mu_2$

Test statistic, $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2(\frac{1}{n_1} + \frac{1}{n_2})}}$.....(1)

$\bar{x}_1 = \frac{\sum x_{1i}}{n_1} = 114/7 = 16.29$

$\bar{x}_2 = \frac{\sum x_{2i}}{n_2} = 163/11 = 14.82$

$s_1^2 = \frac{1}{n_1 - 1}[\sum x_{1i}^2 - \frac{(\sum x_{1i})^2}{n_1}] = 1/6(1898 - 114^2/7) = 6.905$

$s_2^2 = \frac{1}{n_2 - 1}[\sum x_{2i}^2 - \frac{(\sum x_{2i})^2}{n_2}] = 1/10(2569 - 163^2/11) = 15.364$

Pooled variance, $s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{6(6.905) + 10(15.364)}{7 + 11 - 2} = 12.912$

From equation (1), $t = \frac{16.29 - 14.82}{\sqrt{12.912(\frac{1}{7} + \frac{1}{11})}} = 0.87$

$t_{cal} = |t| = 0.87$ and $t_{tab} = t_{n_1 + n_2 - 2} = t_1 6 = 2.12$ [from t-table with 5% level of significance]

$t_{cal} < t_{tab}$, so we cannot reject H_0 and the employment facility for both the universities is similar.

8.1.1.3 Test of equality of several means

Example 3: There are 25 computers in an office supplied by 5 companies in lots of 5. The failure time of different computers are shown below:

Company	Failure time (in hour) of computers				
A	1.5	1	0.5	2	3
B	1	0.5	0.2	0.5	0.5
C	0	2	2	0	0
D	2	2.5	1	2	3
E	3	3	2	2.5	2.5

(a) Are the computers of different companies alike in failure time?

(b) Is there any difference between computers of company B and company E?

Solution (a):

We have to test, Null hypothesis, $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$

Alternative hypothesis, H_1 :At least one of the equality does not hold good

Here, $n_1 = n_2 = n_3 = n_4 = n_5 = 5$, $n = 25$

Company	Failure time (in hour) of computers (y_{ij})					Total (y_i)	Mean ($\bar{y}_i$)
A	1.5	1	0.5	2	3	8	1.6
B	1	0.5	0.2	0.5	0.5	2.7	0.54
C	0	2	2	0	0	4	0.8
D	2	2.5	1	2	3	10.5	2.1
E	3	3	2	2.5	2.5	13	2.6

Correction Term, CT= $\frac{G^2}{n} = \frac{38.2^2}{25} = 58.3696$ [G=Grand total]

Total Sum of Squares, TSS=$\sum\sum y_{ij}^2$-CT=$(1.5^2 + 1^2 + 0.5^2 + \ldots + 2.5^2 + 2.5^2) - 58.3696 = 85.04 - 58.3696 = 26.6704$

Company Sum of Squares, SS(Company)=$\frac{\sum y_i^2}{n_i}$−CT=$\frac{1}{5}(8^2 + 2.7^2 + 4^2 + 10.5^2 + 13^2) - 58.369 = (366.54/5) - 58.3696 = 14.9384$

Error Sum of Squares, SS(Error)=TSS-SS(Company)=$26.6704 - 14.9384 = 11.732$

Table 8.1: ANOVA Table

Sources of variation	Degrees of freedom (df)	Sum of Squares (SS)	Mean Sum of Squares (MS=SS/df)	$F_{cal} = \frac{MS_{Company}}{MS_{Error}}$	$F_{tab} = F_{k-1,N-k}$
Company	k-1=5-1=4	14.9384	14.9384/4=3.7346		
Error	N-k=25-5=20	11.732	11.732/20=0.5866	F=3.7346/0.5866=6.37	$F_{4,20} = 2.87$
Total	N-1=25-1=24	26.6704			

$F_{cal} > F_{tab}$, H_0 is rejected. So, the computers of different companies aren't similar in failure time.

Solution (b):

We have to test, $H_0 : \mu_2 = \mu_5$ vs $H_1 : \mu_2 \neq \mu_5$

Test statistic, $t = \frac{\bar{y}_2 - \bar{y}_5}{\sqrt{\frac{2 \times MS_{Error}}{n_i}}} = \frac{0.54 - 2.6}{\sqrt{\frac{2 \times 0.5866}{5}}} = -4.25$

$t_{cal} = |t| = 4.25$ & $t_{tab} = t_{Error(df)} = t_{0.05,20} = 2.086$

$t_{cal} > t_{tab}$

So H_0 is rejected and there is difference between computers of company B and company E.

8.1.2 Test regarding Proportions

Hypothesis testing applications with a dichotomous outcome variable in a single population are also performed. Similar to tests for means, a key component is setting up the null and research hypotheses. The objective is to compare the proportion of successes in a single population to a known proportion (p0). That known proportion is generally derived from another study or report and is sometimes called a historical control. It is important in setting up the hypotheses in a one sample test that the proportion specified in the null hypothesis is a fair and reasonable comparator. Specifically, we compute the sample size (n) and the sample proportion which is computed by taking the ratio of the number of successes(a) to the sample size,

Proportion, $P = \frac{a}{n}$

8.1.2.1 Test regarding one proportion

Example 4:

Students	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Grade	B	C	A	D	C	C	D	B	B	A	C	D	D	C	B

Do you think that 10% students get grade A?

Solution:

a=2, n=15, proportion, P=a/n=2/15=0.13

We need to test, $H_0 : P = P_0 = 0.10$ vs $H_1 : p \neq 0.10$

Test statistic, $z = \frac{P - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} = \frac{0.13 - 0.10}{\sqrt{\frac{0.10 \times 0.90}{15}}} = 0.39$ $[Q_0 = 1 - P_0 = 1 - 0.10 = 0.90]$

$z_{cal} = 0.39$ & $z_{tab} = z_{0.05} = 1.96$

$z_{cal} < z_{tab}$, so H_0 cannot be rejected and we can conclude that 10% students get grade A.

8.1.2.2 Test regarding two proportions

Example 5: Is the severity of the drug problem in high school the same for boys and girls? 85 boys and 70 girls were questioned and 34 of the boys and 14 of the girls admitted to having tried some sort of drug.

Solution:

We need to test the null hypothesis, $H_0 : P_1 = P_2$

Alternative hypothesis, $H_1 : P_1 \neq P_2$

Test statistics, $z = \frac{p_1 - p_2}{\sqrt{PQ(\frac{1}{n_1} + \frac{1}{n_2})}}$.....(1)

Here, Proportion of boys who tried drug, $p_1 = a_1/n_1 = 34/85 = 0.4$

Proportion of girls who tried drug, $p_2 = a_2/n_2 = 14/70 = 0.2$

$P = \frac{a_1 + a_2}{n_1 + n_2} = \frac{34+14}{85+70} = \frac{48}{155} = 0.31$

$Q = 1 - P = 1 - 0.31 = 0.69$

From equation (1), $z = \frac{0.4 - 0.2}{\sqrt{0.31 \times 0.69(\frac{1}{85} + \frac{1}{70})}} = 2.68$

$z_{cal} = 2.68$ & $z_{tab} = z_{0.05} = 1.96$

$z_{cal} > z_{tab}$

So H_0 is rejected and the severity of the drug problem in high school the aren't same for boys and girls.

8.1.2.3 Test regarding several proportions

Example 6: The following are the number of defective computers of different laboratories:

Laboratories	L_1	L_2	L_3	L_4
No of defective computers	8	15	5	12

Are the proportions of defective computers of different laboratories similar?

Solution:

Let P_1, P_2, P_3 and P_4 are the proportions of defective computers.

We need to test $H_0 : P_1 = P_2 = P_3 = P_4$ vs H_1 : At least one of the equality does not hold good.

Here, $n = 8 + 15 + 5 + 12 = 40$ & $k = 4$(no of laboratories)

Under null hypothesis, Expected frequency, $E_i = \frac{n}{k} = 40/4 = 10$

Test statistics, chi-square(χ^2)=$\sum \frac{O_i^2}{E_i} - n = \frac{1}{10}(8^2 + 15^2 + 5^2 + 12^2) - 40 = 45.8 - 40 = 5.8$

$\chi^2_{cal} = 5.8$ & $\chi^2_{tab} = \chi^2_{0.05,k-1} = \chi^2_{0.05,3} = 7.81$

$\chi^2_{cal} < \chi^2_{tab}$

So H_0 can not be rejected and the proportions of defective computers of different laboratories are similar.

8.2 Test of independence or, Test of association

Let us consider that a researcher has n units in a sample. The sample observations are classified according to qualitative characters, say A and B as follows:

Table 8.2: Cross table of A & B

A B	B	Not B	Total
A	O_{11} = a	O_{12} = b	R_1 = a+b
Not A	O_{21} = c	O_{22} = d	R_2 = c+d
Total	C_1 = a+c	C_2 = b+d	$n = R_1 + R_2 + C_1 + C_2$

Here, a, b, c, d are observed frequencies in different cells.

$n = \sum\sum O_{ij}$.

O_{ij} = observation of ith row and j^{th} column recorded from the experiment.

$E_{ij} = \frac{R_i C_j}{n}$ = expected frequency corresponding to i^{th} row and j^{th} column under H_0.

Hypothesis:

H_0 : The characters A and B are independent or, There is no association between A and B.

H_1 : They are not independent or, There is association between A and B.

Test statistic:

Chi-square statistic, $\chi^2 = \sum\sum \frac{O_{ij}^2}{E_{ij}} - n$.

It has $(r-1)(c-1)$ degrees of freedom (df). Here, r = no. of rows and c = no. of columns.

For $r = 2$ and $c = 2$,

$$\chi^2 = \frac{n(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} \sim \chi_1^2$$

Example 7:

150 computer graduates are interviewed and are classified according to their result and job satisfaction. The numbers of graduates in different classes are:

Table 8.3: Cross table of Results and Job Satisfaction

Result	Job Satisfaction		Total
	Yes	No	
Good	22	58	80
Not Good	20	50	70
Total	42	108	150

Do you think that the graduates with good result are satisfied with their job?

Solution:

H_0 : They are independent i.e, Job satisfaction does not depend on good result.

H_1 : They are not independent i.e, The good result and job satisfaction are associated.

The test statistic is:

$$\begin{aligned}\chi^2 &= \frac{n(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} \\ &= \frac{150(22\times 50 - 20\times 58)^2}{80\times 70\times 42\times 108} \\ &= 0.02\end{aligned}$$

Here, $\chi^2_{tab} = \chi^2_{0.05,1} = 3.84$

$\chi^2_{cal} = 0.02$

$\chi^2_{cal} < \chi^2_{tab}$.

Hence, H_0 is accepted.

Example 8:

The following are number of emails of different organizations, where emails are classified according to local and foreign emails. The classified results are shown below:

Table 8.4: Cross table of Origin of emails and Emails of organizations

Origin of emails	Emails of organizations			Total
	O_1	O_2	O_3	
Local	45	25	12	82
Foreign	08	12	18	38
Total	53	37	30	120

Is there any association between origin of emails and organizations?

Solution:

We need to test,

H_0 :They are independent or, Origin of emails does not depend on organization

H_1 :They aren't independent or, Origin of emails depends on organization.

Expected frequency corresponding to i^{th} row and j^{th} column under H_0 is E_{ij} and $E_{ij} = \frac{R_i C_j}{n}$.

Here, $n = 120$.

For observed value 45, row total $R_1 = 82$, column total $C_1 = 53$ and $E_{11} = \frac{R_1 C_1}{n} = \frac{82 \times 53}{120} = 36.22$.

For observed value 18, row total $R_2 = 38$, column total $C_3 = 30$ and $E_{23} = \frac{R_2 C_3}{n} = \frac{38 \times 30}{120} = 9.5$.

Table 8.5: Cross table of Origin of emails and Expected frequency

Origin of emails	Expected frequency			Total
	E_1	E_2	E_3	
Local	$\frac{82\times53}{120} = 36.22$	$\frac{82\times37}{120} = 25.28$	$\frac{82\times30}{120} = 20.5$	82
Foreign	$\frac{38\times53}{120} = 16.78$	$\frac{38\times37}{120} = 11.72$	$\frac{38\times30}{120} = 9.5$	38
Total	53	37	30	120

The test statistic is,

$$\begin{aligned}\chi^2 &= \sum\sum \frac{O_{ij}^2}{E_{ij}} - n \\ &= (\frac{45^2}{36.22} + \frac{25^2}{25.28} + \frac{12^2}{20.5} + \frac{8^2}{16.78} + \frac{12^2}{11.72} + \frac{18^2}{9.5}) - 120 \\ &= 137.89 - 120 \\ &= 17.89\end{aligned}$$

Here, $\chi^2_{tab} = \chi^2_{0.05,(r-1)(c-1)} = \chi^2_{0.05,(2-1)(3-1)} = \chi^2_{0.05,2} = 5.99$

$\chi^2_{cal} = 17.89$

$\chi^2_{cal} > \chi^2_{tab}$.

Hence, H_0 is rejected.

8.2.1 Exercise:

1. Write down examples of parameter, statistic, null and alternative hypotheses.
2. Given, the no. of dissatisfied customers of a mobile operator in different days: No. of dissatisfied customers(x): 2, 6, 3, 10, 11, 12.

   Do you think that the average number of dissatisfied customer is 8?
3. The following are the no. of emails sent from two cyber cafs in different days:

   Emails sent from cyber cafe-1(x_{1j}): 12, 18, 17, 20, 15, 10, 20

   Emails sent from cyber cafe-2(x_{2j}): 15, 10, 25, 17, 16, 20

   Are the two-cyber caf similar in sending no. of emails?
4. Are the organizations similar in respect of receiving emails?

Company	Emails received in different days				
O_1	10	8	12	9	8
O_2	7	15	12	8	10
O_3	6	10	12	10	9
O_4	12	11	10	8	5

5. The information provided from a sample of 40 observations are: $\sum x = 673$ and $\sum x^2 = 11385$. Is the sample selected from a population with mean 18?

6. In a laboratory, there are 18 Samsung and 20 Acer computers. In another laboratory, there are 12 Samsung and 14 Acer computers. Are the proportions of Samsung computers in both laboratories similar?

7. The following data are related to some graduates of different subjects who are satisfied with their job status.

Table 8.6: Cross table of Subjects and Satisfaction

Satisfaction	Subjects				Total
	BA	EEE	CS	CSE	
Yes	25	10	11	30	76
No	12	13	10	9	44
Total	37	23	21	39	120

Is there any association between subjects taught and job satisfaction?

8. A group of students are classified by their residential origin and full attention to learning:

Table 8.7: Cross table of Residential origin and Full attention

Residential origin	Full attention		Total
	Yes	No	
Rural	28	32	60
Urban	44	70	114
Total	72	102	174

Is full attention to learning associated with residential origin?

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

t-table.xls 7/14/2007

Figure 8.1: t-table

Table of critical values for the F distribution (for use with ANOVA):

How to use this table:

There are two tables here. The first one gives critical values of F at the p = 0.05 level of significance.
The second table gives critical values of F at the p = 0.01 level of significance.
1. Obtain your F-ratio. This has (x,y) degrees of freedom associated with it.
2. Go along x columns, and down y rows. The point of intersection is your critical F-ratio.
3. If your obtained value of F is equal to or larger than this critical F-value, then your result is significant at that level of probability.
An example: I obtain an F ratio of 3.96 with (2, 24) degrees of freedom.
I go along 2 columns and down 24 rows. The critical value of F is 3.40. My obtained F-ratio is larger than this, and so I conclude that my obtained F-ratio is likely to occur by chance with a p<.05.

Critical values of F for the 0.05 significance level:

	1	2	3	4	5	6	7	8	9	10
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.39	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
10	4.97	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
11	4.84	3.98	3.59	3.36	3.20	3.10	3.01	2.95	2.90	2.85
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
17	4.45	3.59	3.20	2.97	2.81	2.70	2.61	2.55	2.49	2.45
18	4.41	3.56	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
21	4.33	3.47	3.07	2.84	2.69	2.57	2.49	2.42	2.37	2.32
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.38	2.32	2.28
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.26
25	4.24	3.39	2.99	2.76	2.60	2.49	2.41	2.34	2.28	2.24
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.17
31	4.16	3.31	2.91	2.68	2.52	2.41	2.32	2.26	2.20	2.15
32	4.15	3.30	2.90	2.67	2.51	2.40	2.31	2.24	2.19	2.14
33	4.14	3.29	2.89	2.66	2.50	2.39	2.30	2.24	2.18	2.13
34	4.13	3.28	2.88	2.65	2.49	2.38	2.29	2.23	2.17	2.12
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22	2.16	2.11

Figure 8.2: F-table

Chi-square Distribution Table

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89
32	15.13	16.36	18.29	20.07	22.27	42.58	46.19	49.48	53.49
34	16.50	17.79	19.81	21.66	23.95	44.90	48.60	51.97	56.06
38	19.29	20.69	22.88	24.88	27.34	49.51	53.38	56.90	61.16
42	22.14	23.65	26.00	28.14	30.77	54.09	58.12	61.78	66.21
46	25.04	26.66	29.16	31.44	34.22	58.64	62.83	66.62	71.20
50	27.99	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15
55	31.73	33.57	36.40	38.96	42.06	68.80	73.31	77.38	82.29
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38
65	39.38	41.44	44.60	47.45	50.88	79.97	84.82	89.18	94.42
70	43.28	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.43
75	47.21	49.48	52.94	56.05	59.79	91.06	96.22	100.84	106.39
80	51.17	53.54	57.15	60.39	64.28	96.58	101.88	106.63	112.33
85	55.17	57.63	61.39	64.75	68.78	102.08	107.52	112.39	118.24
90	59.20	61.75	65.65	69.13	73.29	107.57	113.15	118.14	124.12
95	63.25	65.90	69.92	73.52	77.82	113.04	118.75	123.86	129.97
100	67.33	70.06	74.22	77.93	82.36	118.50	124.34	129.56	135.81

Figure 8.3: Chi-square table